

MSc. Thesis Phase 1

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Part 1&2

Research Foundations

The thesis represents more than just a graduation requirement; it is your opportunity to demonstrate **mastery of research methodologies, technical expertise, critical thinking, and academic communication**.

Whether you are **exploring machine learning applications** in healthcare, analyzing social media data for policy insights, or developing predictive models for agricultural sustainability, your work will reflect both rigorous scientific inquiry and practical relevance to society.

Learning Objectives

Established a solid foundation in research fundamentals and **begun the critical process of topic identification**.

These objectives are designed to ensure you understand not just what you need to do, but why these **practices matter for producing ethical, rigorous, and impactful research**.



Understand Thesis Structure

Grasp the complete architecture of an MSc thesis, including goals, structural components, key milestones, and assessment criteria. Learn how proposals, literature reviews, methodology chapters, results, and discussions interconnect to tell a coherent research story.



Differentiate Research Quality

Develop critical evaluation skills to distinguish between rigorous, well-designed research and methodologically flawed or ethically problematic work. Recognize hallmarks of excellence in data science research and common pitfalls to avoid.



Apply Ethics & Integrity

Master fundamental principles of research ethics, academic integrity, data governance, and proper citation practices. Understand how ethical considerations shape every stage of research from data collection through dissemination.



Generate Research Topics

Develop and articulate 2–3 potential research topics that are feasible, data-driven, academically valuable, and personally meaningful. Learn systematic approaches to identifying problems worth solving through data science.

Welcome

What is a Data Science MSc Thesis?

Your MSc thesis is an **independent research project** that demonstrates **your ability to identify a meaningful problem, design and execute a rigorous investigation** using **data science methods, analyze results critically, and communicate findings effectively**. Unlike coursework assignments where problems and datasets are provided, the *thesis* requires you to formulate **research questions, source or generate data, select appropriate analytical approaches, and contribute original insights to the field**.

Understanding the Distinctions

- **Research Thesis:** Emphasizes **novel contributions** to knowledge, **methodological innovation**, and **theoretical frameworks**. Typically involves **extensive literature** review and positions work within existing scholarly conversations.
- **Applied Project:** Focuses on **solving practical problems using established methods**, often in partnership with industry or community organizations. Emphasizes implementation and real-world impact.
- **Capstone:** Demonstrates comprehensive mastery of program competencies through a substantial project that integrates multiple areas of learning.

Key Thesis Deliverables

Throughout your thesis journey, you'll produce several critical documents and presentations that mark your progress and demonstrate growing expertise:

1. **Research Proposal:** A detailed plan outlining your research question, literature foundation, methodology, timeline, and expected contributions. This document receives formal approval before you proceed.
2. **Progress Reviews:** Regular checkpoints where you present completed work, address challenges, and receive guidance from your supervisor and potentially a review committee.
3. **Final Thesis Document:** The comprehensive written report documenting your entire research process, findings, and conclusions according to department formatting standards.
4. **Thesis Defense:** An oral examination where you present your research and respond to questions from an examination committee, demonstrating deep understanding of your work.

Roles and Responsibilities: A Partnership Approach

Success in your thesis requires understanding that research is a collaborative yet independent endeavor. While you will receive **guidance and support**, you are the primary driver of your research project. Clarity about roles prevents misunderstandings and helps establish productive working relationships.

	Student Responsibilities
	Initiative and Ownership: You must proactively manage your project, set intermediate goals, troubleshoot problems, and maintain momentum even when facing setbacks. Research requires self-direction and intrinsic motivation.
	Meeting Deadlines: Respect supervisor time and institutional requirements by submitting work on schedule. Late submissions create cascading problems and may jeopardize your graduation timeline.
	Academic Integrity: Maintain the highest ethical standards in all aspects of your work, from data collection through writing. Your reputation and career depend on trustworthiness.
	Communication: Keep your supervisor informed about progress, challenges, and changes in direction. Don't disappear for weeks, regular updates build confidence and enable timely support.
	Supervisor's Role
	Strategic Guidance: Your supervisor provides expert advice on research design, methodological choices, literature interpretation, and manuscript development. They help you navigate academic conventions and avoid common pitfalls.
	Quality Assurance: Supervisors review your work critically, identifying weaknesses, suggesting improvements, and ensuring your research meets academic standards for rigor and contribution.
	Not "Doing the Work": Supervisors guide but don't conduct your research. They won't debug your code, write your chapters, or solve your analytical problems, they help you develop the skills to handle these yourself.
	Professional Development: Beyond the thesis, good supervisors help you develop broader research competencies, connect with academic networks, and prepare for career transitions.

Effective Communication Practices

Establish clear communication protocols early: How often will you meet? What format should progress updates take? How quickly should you expect feedback? Consider weekly written updates summarizing accomplishments, challenges, and next steps, supplemented by bi-weekly or monthly face-to-face meetings for deeper discussion. Come to meetings prepared with specific questions rather than vague requests for help. Document decisions and action items to ensure accountability.

Research Ethics and Integrity in Data Science

Data science research carries unique ethical responsibilities because our work directly affects individuals, communities, and societies. Unlike purely theoretical research, data science projects often involve personal information, can perpetuate biases, and may influence consequential decisions in domains like healthcare, criminal justice, employment, and finance. Understanding ethical principles isn't just about avoiding problems, it is about conducting research that genuinely serves the public good.

Core Ethical Principles in Data Science Research

Beneficence and Non-Maleficence: Research should aim to benefit society while minimizing potential harms. Before deploying a predictive model or publishing findings, consider: Who might be helped? Who might be harmed? How can negative impacts be mitigated?

Justice and Fairness: Research benefits and burdens should be distributed equitably. Be particularly vigilant about algorithmic bias that might disadvantage already marginalized groups. Ensure diverse representation in training data and validate model performance across demographic subgroups.

Respect for Persons: Individuals have the right to make informed decisions about participating in research and to have their privacy protected. This principle underlies requirements for informed consent and data protection.

Transparency and Accountability: Research methods, limitations, and potential conflicts of interest should be clearly disclosed. Open science practices, sharing code, data, and detailed methodology, enhance reproducibility and public trust.

Common Ethical Challenges

- Using data collected for one purpose in new contexts without consent
- Re-identification risks when "anonymized" datasets contain quasi-identifiers
- Training models on biased historical data that perpetuate discrimination
- Inadequate security measures leading to data breaches
- Publishing findings that could enable harmful applications

📋 **Institutional Review Board (IRB) / Ethics Committee Approval:** If your research involves human subjects, including analyzing social media data, conducting surveys, or studying health records, you must obtain ethics approval before beginning data collection. The review process typically takes 1-4 weeks, so plan accordingly. Your supervisor can guide you through this process.

Plagiarism And Academic Writing Integrity

Academic integrity forms the foundation of scholarly trust. When researchers plagiarize, whether intentionally or through carelessness, they undermine the entire knowledge ecosystem. In an era of AI writing assistants and easily accessible online content, understanding what constitutes original work and proper attribution has never been more important.



Understanding Plagiarism

Plagiarism includes copying text without quotation marks and citation, paraphrasing someone's ideas without attribution, reusing your own previous work without disclosure (self-plagiarism), and submitting work done by others as your own. Even unintentional plagiarism carries serious consequences.



Ethical AI Tool Use

AI assistants like ChatGPT, Copilot, and Grammarly can support your writing, but using them requires transparency and judgment. Acceptable uses include brainstorming ideas, improving grammar, and explaining concepts. Unacceptable uses include generating entire sections without disclosure or substituting AI output for your own analysis.



Proper Citation Practices

Every idea, fact, or quote from another source requires attribution. Use reference management tools like **Mendeley** or **Zotero** to organize sources and generate consistent citations. Learn your field's preferred style (APA, IEEE, Chicago) and apply it meticulously throughout your thesis.



Similarity Detection

Tools like **Turnitin** compare your writing against billions of sources to identify matches. High similarity scores don't automatically indicate plagiarism, properly cited quotes and common phrases appear similar, but require careful review. Aim for originality in expression while acknowledging intellectual debts.

Best Practices for Academic Integrity

- Take detailed notes during literature review, clearly distinguishing your ideas from sources
- Cite as you write rather than adding references later when memories blur
- When in doubt about whether citation is needed, cite—over-attribution is better than under-attribution
- Disclose any AI assistance in your methods section or acknowledgments
- Understand that your thesis should reflect YOUR thinking, analysis, and voice—tools should enhance, not replace, your intellectual work

Identifying a Research-Worthy Topic

Perhaps the most anxiety-inducing aspect of beginning a thesis is choosing a topic. Students often struggle with topics that are too broad ("machine learning in healthcare"), too narrow ("predicting tomorrow's weather in my Accra"), or insufficiently grounded in available data. A research-worthy topic balances **ambition** with **feasibility**, **passion** with **practicality**, and **novelty** with relevance.

The Three Pillars of Research-Worthiness

Novelty And Academic Contribution

Your research should extend existing knowledge in some way, whether by **applying methods** in new contexts, **addressing previously** unstudied questions, **challenging existing assumptions**, or **improving upon** prior approaches. This doesn't mean everything must be revolutionary; incremental advances matter.

Review recent literature to identify **gaps**, **contradictions**, or **opportunities** for enhancement.

Practical Value And Real-World Impact

The best data science research addresses problems that matter to people. Consider: Who faces this challenge? How do current solutions fall short? What would improved understanding or methods enable? Topics connecting to health, sustainability, education, economic development, or social justice often offer rich opportunities for meaningful impact.

Data Feasibility And Technical Viability

Brilliant research questions become impossible projects without accessible data of sufficient quality and quantity. Before committing to a topic, verify: Does relevant data exist? Can you legally and ethically access it? Is it structured appropriately for analysis? Do you have or can you develop the technical skills needed? A six-month MSc timeline demands pragmatic scoping.

The "3W" Model for Topic Evaluation

Apply this simple framework to any potential topic:

- **What is the problem?** Can you articulate the specific issue in 2-3 clear sentences? Is it well-defined enough to study systematically?
- **Why does it matter?** Who is affected and how? What are the consequences of leaving it unsolved? Why should anyone care about your findings?
- **Who benefits?** How will your research create value, for academics, practitioners, communities, or policymakers? What actionable insights might it generate?

Sources of Research Ideas

Research ideas don't emerge from thin air, they come from **systematic exploration of problems, literature, and available resources**. Cultivating curiosity and developing a researcher's eye for opportunity transforms everyday observations into potential thesis topics.

Published Academic Literature

Read recent papers in top-tier venues (IEEE, ACM, Elsevier journals, NeurIPS, ICML) to understand current research frontiers. Pay special attention to **"Future Work"** sections where authors identify limitations and extensions.

Look for:

- Methods tested in one domain that could apply to another
- Conflicting findings requiring reconciliation
- Techniques emerging in computer science but not yet applied to your interest area
- Datasets or benchmarks in need of updating

Industry And Open Data Challenges

Real-world problems provide excellent thesis foundations. Explore:

- **Kaggle competitions:** Current and past challenges reveal industry priorities and provide ready-made datasets
- **GitHub repositories:** Open source projects often document unsolved problems and improvement opportunities
- **Government open data portals:** UN, World Bank, national statistical agencies, and city data repositories offer rich, policy-relevant datasets
- **Company partnerships:** Some supervisors maintain industry connections enabling applied research projects

Local And Regional Challenges

For students in Africa and other LMICs (Low and Middle Income Countries), local context offers unique research opportunities often underexplored in mainstream data science:

- Agricultural productivity and climate adaptation
- Mobile money and financial inclusion
- Healthcare delivery in resource-constrained settings
- Education access and quality improvement
- Infrastructure and urbanization challenges

These topics not only offer research contributions but also position your work to create tangible social impact.

Brainstorming Framework: From Interest to Viable Topic

Follow this systematic progression: **Personal Interests → Domain Exploration → Data Discovery → Problem Refinement → Feasibility Check**. Start with broad areas you are passionate about (healthcare, environment, education), then narrow focus as you learn about specific challenges, data availability, and methodological approaches. Document this process, your early explorations often provide valuable context for your thesis introduction.

Class Activity: Topic Discovery Canvas

This structured brainstorming activity guides you from vague interests to concrete, feasible research topics. Work through each step methodically, and don't rush, quality topics emerge from careful consideration and iteration.

01

List Personal Interests

Identify 3-5 domains that genuinely excite you. Think broadly: healthcare, climate change, education technology, urban development, financial systems, social networks, agriculture, sports analytics, entertainment, governance. Your thesis will consume significant time and energy, passion sustains effort through inevitable challenges.

Consider: What problems keep you reading past midnight? What issues do you discuss enthusiastically with friends?

03

Frame Potential Research Questions

Transform interests and data into specific, answerable questions. Use question starters: "Can we predict...?", "How do X and Y relate...?", "What patterns exist in...?", "How effective is X for...?", "What factors influence...?"

Example transformations: Interest (healthcare) + Data (hospital admissions) → Question ("Can we predict patient readmission risk using admission patterns and comorbidity data?"). **Aim for questions requiring analysis rather than simple description.**

02

Identify Available Datasets

For each interest area, research publicly accessible data sources. **Explore:** government open data portals, academic repositories (UCI Machine Learning Repository, Kaggle, Google Dataset Search), API-accessible data (Twitter, weather services, financial markets), research collaborations, or institutional data partnerships. Document dataset characteristics: size, variables, temporal coverage, licensing terms, and known limitations.

04

Peer Feedback & Refinement

Present your draft topics to peers and receive constructive feedback. **Ask:** Is the question clear? Does it seem feasible in scope? Is the value proposition evident? What challenges or opportunities do they see? Use this input to refine and strengthen your topics. Select 2-3 finalists that you'll explore more deeply through preliminary literature review and supervisor consultation. Document your decision rationale, why these topics among alternatives?

📄 **Activity Output:** By session end, every student should have a completed Topic Discovery Canvas/Doc with 2-3 well-articulated potential topics, each including a tentative research question, data source identification, preliminary value proposition, and acknowledged challenges. This document becomes the foundation for supervisor discussions and problem statement development.

Key Resources

Essential Readings for Research Foundations

These texts provide comprehensive guidance on research design, methodology selection, and academic writing. You don't need to read them cover-to-cover immediately, but familiarize yourself with their structure so you can reference relevant sections as needs arise throughout your thesis journey.

- **Creswell, J. W. (2018). Research Design: Qualitative, Quantitative, and Mixed Methods Approaches (5th ed.).** Sage Publications. — The definitive guide to research methodology across paradigms. Chapters 1-3 establish foundational concepts applicable to all research. Data science students particularly benefit from chapters on quantitative and mixed methods designs.
- **Hair, J., Page, M., & Brunsveld, N. (2021). Essentials of Business Research Methods (4th ed.).** Routledge. Practical, accessible introduction to research processes with emphasis on applied projects. Excellent sections on problem definition, data collection, and presenting findings.
- **Your University's Thesis Handbook (Department-Specific).** — This document specifies formatting requirements, submission procedures, examination processes, and institutional policies. Review it carefully to avoid last-minute formatting crises.
- **Research Ethics Training Modules:** Complete online courses through CITI Program (citiprogram.org) or Coursera's "Data Science Ethics" specialization. Many institutions require certification before ethics approval.



Additional Online Resources

- [Google Scholar \(scholar.google.com\)](https://scholar.google.com) — Search academic literature
- [arXiv.org](https://arxiv.org) — Preprints in computer science and statistics
- [Papers With Code \(paperswithcode.com\)](https://paperswithcode.com) — Research papers linked to implementation code
- [University thesis repositories](#) — Examples from previous MSc graduates in your program

PART 2

Topic Selection And Problem Identification

These section represents a critical transition from exploration to commitment. Having generated multiple potential topics in part 1, you now begin the rigorous process of evaluation, refinement, and selection. This week emphasizes transforming a broad topic into a precise, researchable problem, arguably the most important step in the thesis process. A well-defined problem provides clarity, focus, and direction for all subsequent work.

Learning Objectives

1

Refine and Justify One Viable Topic

Through critical evaluation against feasibility, novelty, and impact criteria, select a single topic that will become your thesis focus. Develop compelling justification for why this problem deserves investigation and why you are positioned to study it effectively.

2

Frame a Clear Problem Statement

Articulate your research problem with precision, specifying what exactly you will investigate, why it matters, and what you aim to achieve. Learn to align problem statements with data science methodological approaches, ensuring your question matches available methods.

3

Establish Relevance

Demonstrate both academic significance (how your research contributes to scholarly knowledge) and societal impact (how findings could benefit practitioners, communities, or policymakers). Strong research sits at the intersection of theoretical advancement and practical utility.

4

Engage in Supervisor Consultation

Prepare for and conduct productive meetings with your supervisor, asking targeted questions that advance your thinking. Learn to synthesize feedback, defend your choices with evidence, and iterate your problem definition based on expert guidance.

From Idea to Research Problem

Students frequently confuse *topics* (broad areas of interest) *with research problems* (specific, bounded questions). "Healthcare analytics" is a topic encompassing thousands of potential problems. "Predicting hospital readmission risk for heart failure patients using electronic health record data" is a research problem, specific, measurable, and actionable. This distinction matters because research requires focus. You cannot study everything about a topic in one thesis; you must identify a slice worth deep investigation.

Characteristics of a Well-Defined Research Problem

Specific: The problem clearly delineates scope, what's included and excluded. Vague: "Social media and health."
Specific: "Using Twitter sentiment analysis to detect regional influenza outbreaks in sub-Saharan Africa."

Measurable: You can determine when the problem is solved or partially addressed through empirical evidence. What outcomes, metrics, or evaluation criteria indicate success?

Achievable: Within your timeline, resources, skills, and data access, the problem is solvable. Ambition is valuable, but overreach leads to frustration and incomplete work.

Relevant: The problem matters to some audience—academics, practitioners, policymakers, or communities. It addresses a genuine gap or need rather than an arbitrary question.

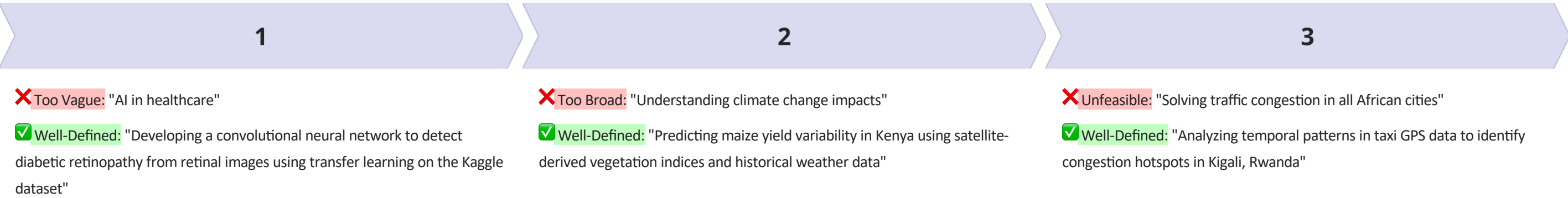
Time-bound: You can complete the investigation within your program timeline. This often means accepting that your thesis will be a stepping stone toward larger questions rather than definitive answers to sprawling problems.

Data-Driven: The problem is amenable to investigation through data analysis. Pure speculation, purely qualitative questions without data collection plans, or questions requiring unavailable data don't fit data science thesis expectations.

SMART Framework Application

Apply the SMART criteria (Specific, Measurable, Achievable, Relevant, Time-bound) to pressure-test your problem definition. If any dimension fails, refine until all five criteria are solidly met. This exercise forces clarity and prevents scope creep.

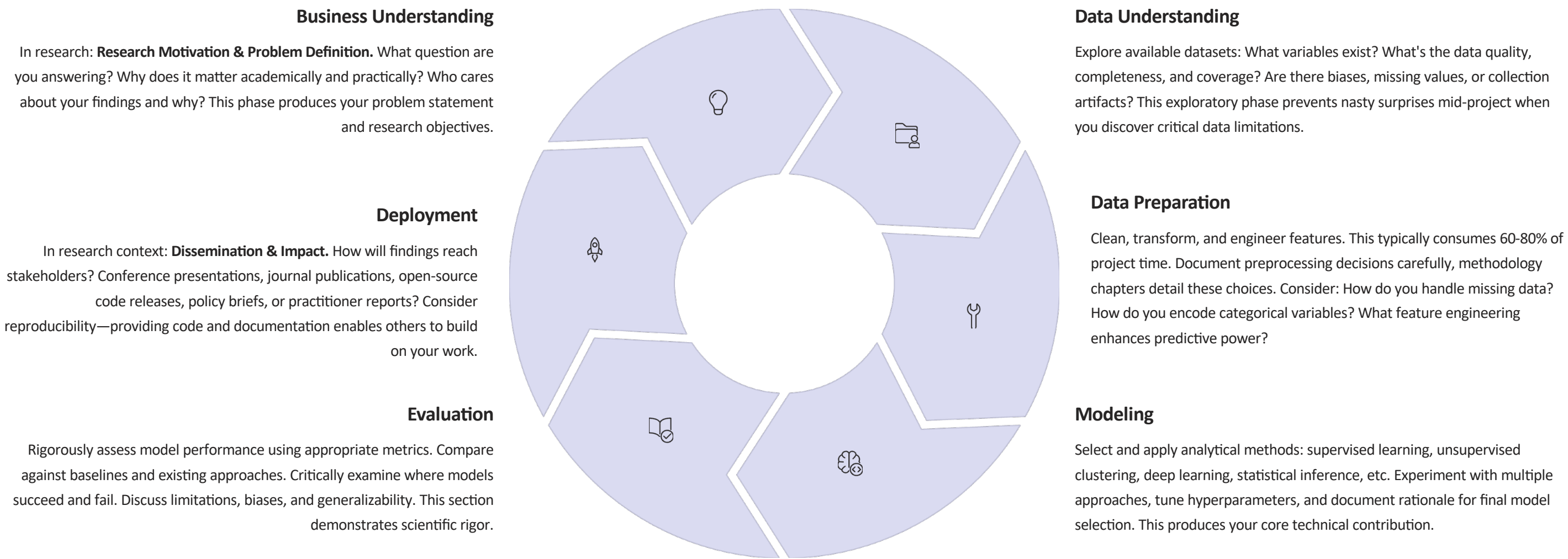
Example Problem Transformations



Problem Framing Using CRISP-DM

The Cross-Industry Standard Process for Data Mining (CRISP-DM) provides an industry-proven framework for structuring data science projects. Though initially developed for commercial analytics, CRISP-DM translates beautifully to academic research, offering a systematic approach to moving from problem to deployed solution. Mapping your thesis to CRISP-DM early ensures you consider all critical dimensions and avoid common pitfalls like beginning modeling before understanding data quality.

CRISP-DM Phases Applied to Research



CRISP-DM is iterative, not linear. You will cycle between phases as you refine understanding. For example, modeling attempts might reveal data quality issues requiring return to preparation, or evaluation results might prompt reframing research questions (business understanding). Embracing this iteration prevents rigid adherence to flawed plans.

Balancing Academic and Industry Relevance

The most impactful research bridges academic rigor with real-world applicability. Pure academic contributions advance knowledge but may lack immediate utility. Pure applied work solves immediate problems but may not generalize or advance methodological innovation. Your thesis should thoughtfully position itself in this landscape, ideally finding the sweet spot where scholarly contribution meets practical value.

Three Lenses on Research Value

Academic Gap

What haven't scholars studied yet? Where do contradictions, limitations, or unanswered questions exist in published literature?

Academic contributions include:

- Novel methodologies or algorithms
- Applications in previously unstudied contexts
- Challenges to prevailing theories or assumptions
- Improved understanding of phenomena
- Better benchmarks or evaluation frameworks

Strong theses position themselves within scholarly conversations, showing how your work extends, refines, or challenges existing knowledge.

Industry Problem

What challenges do practitioners face? Where do existing solutions fall short in accuracy, efficiency, cost, or accessibility?

Industry relevance includes:

- Operational improvements (faster, cheaper, more accurate)
- New capabilities enabling previously impossible tasks
- Risk reduction or quality enhancement
- Better decision support tools
- Addressing regulatory or compliance needs

Industry-relevant research often leads to partnerships, funding opportunities, and career pathways beyond academia.

Societal Impact

How does your research potentially improve lives, particularly for underserved communities?

Social value includes:

- Health outcome improvements
- Educational access or quality gains
- Economic opportunity creation
- Environmental sustainability
- Social equity and justice advancement

For African students, research addressing local development challenges while contributing methodological innovations offers particularly strong positioning for both academic publication and real-world impact.

Case Examples of Balanced Research

Predictive Policing

Academic: Novel fairness-aware algorithms that reduce demographic bias in crime prediction models. **Industry:** Tools helping law enforcement allocate resources efficiently. **Ethical:** Critical examination of how algorithmic bias perpetuates systemic injustice. This topic bridges machine learning methodology, practical deployment, and crucial ethical considerations.

Crop Yield Forecasting

Academic: Integration of multi-source remote sensing data (satellite imagery, weather, soil) using ensemble methods. **Industry:** Tools enabling farmers and policymakers to anticipate harvests for pricing and food security planning. **Community:** Addresses critical challenge in agricultural economies, particularly relevant in developing regions.

Developing a Problem Statement

Your problem statement is perhaps the most important paragraph you will write, it crystallizes your entire research into a concise, compelling narrative. A well-crafted problem statement guides your work, communicates your purpose to readers, and helps supervisors provide targeted feedback. Spend significant time refining this statement; clarity here prevents confusion later.

Essential Components of a Problem Statement

Context Establish the landscape: What is the broader domain or issue? What do we already know? Why should readers care? This frames your work within a larger conversation, showing it's not an arbitrary question but a logical next step in addressing an important challenge.	Gap Identify what's missing or inadequate: What don't we know? Where do current approaches fall short? What problem persists despite existing efforts? The gap justifies your research, it's the hole your work will fill. Be specific about what's lacking, not just what's interesting.	Aim State your research purpose: What do you intend to achieve? What will you do differently or newly? How will your approach address the identified gap? This is your core research objective, expressed as an action you'll undertake with expected outcomes.
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Problem Statement Formula

A reliable template follows this structure:

"Despite [known fact or existing efforts], [specific problem] persists in [context], leading to [negative impact or missed opportunity]. This research seeks to [your proposed investigation/solution], which will [expected contribution or benefit]."

Example Problem Statements

❏ **Example 1 - Financial Inclusion:** "Despite numerous studies on financial inclusion in developing economies, limited research explores how mobile money transaction patterns can predict credit risk among small and medium enterprises (SMEs) in Rwanda, where 65% of businesses remain unbanked. Traditional credit scoring relies on formal banking history unavailable for most SMEs, perpetuating financial exclusion. This research seeks to develop machine learning models that leverage mobile money behavioral data to assess creditworthiness, potentially enabling financial institutions to extend services to previously unserved entrepreneurs and contributing to economic development goals."

❏ **Example 2 - Healthcare Prediction:** "Although machine learning shows promise in medical diagnosis, most models trained on data from high-income countries perform poorly when deployed in low-resource settings with different disease prevalence, equipment, and patient demographics. This research investigates transfer learning approaches to adapt pneumonia detection models trained on Western chest X-ray datasets for accurate diagnosis in East African hospitals, addressing the critical need for AI-assisted diagnostics that work effectively across diverse clinical contexts and potentially improving healthcare delivery in resource-constrained environments."

Preparing for Supervisor Consultation

Your relationship with your supervisor significantly influences thesis success. Productive meetings don't happen by accident, they require preparation, clear communication, and mutual respect. Supervisors appreciate students who come prepared with specific questions, demonstrate initiative in problem-solving, and integrate feedback systematically. This section provides practical guidance for maximizing the value of supervisory interactions.

Before Your First Meeting: Preparation Checklist

Bring Your Topic Draft

Share your 2-3 potential topics (from Week 1 deliverable) and preliminary problem statement (from Week 2 work) in advance if possible. Don't expect your supervisor to read materials on the spot—send documents 24-48 hours before meetings to allow thoughtful review.

Research Your Supervisor's Expertise

Review their recent publications, research interests, and current projects. Understanding their background helps you ask informed questions and shows respect for their time. If your topic sits outside their primary expertise, acknowledge this and discuss how you'll supplement their guidance (co-supervisor, external expert, specialized readings).

Formulate Specific Questions

Vague questions like "What do you think?" are hard to answer helpfully. Instead, ask targeted questions: "Does this problem seem feasibly scoped for six months?" "Are you aware of recent work in this area I should review?" "Would you recommend methodology X or Y for this type of data?" "What are the biggest risks you see in this approach?"

Document the Conversation

Take detailed notes during meetings—it's easy to forget crucial advice later. After each meeting, write a brief summary of key decisions, action items, and next steps. Share this summary with your supervisor to confirm mutual understanding. This documentation creates accountability and prevents repeated conversations.

Topics for Discussion in Early Meetings

- Topic Feasibility:** Is the scope appropriate? Are data sources viable? Are required methods within your skill level or learnable?
- Literature Pointers:** What seminal papers should you read? Are there recent conferences or journals particularly relevant to your topic?
- Methodological Guidance:** What analytical approaches make sense? What tools or frameworks do they recommend?
- Timeline and Milestones:** What are realistic intermediate deadlines? When should drafts of each chapter be ready?
- Resources and Support:** Are there computing resources, datasets, or expert contacts they can help you access?

Integrating Feedback Effectively

Supervisors offer advice based on experience, but you're ultimately responsible for decisions. When you receive feedback:

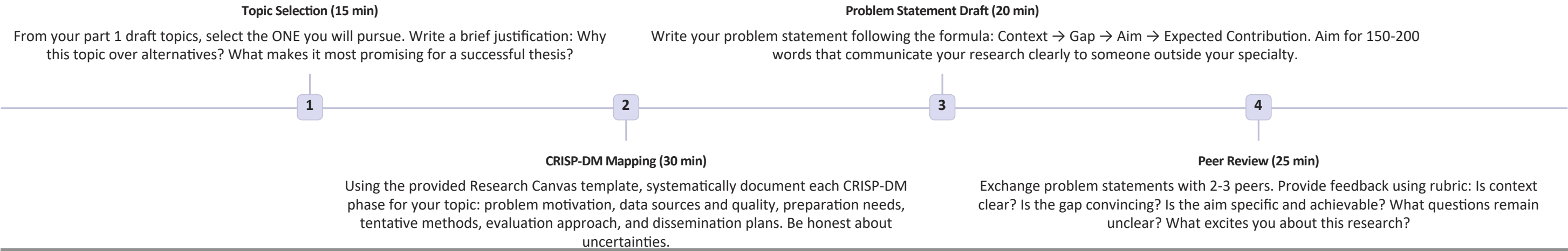
- Clarify Understanding:** If suggestions are unclear, ask follow-up questions immediately rather than making assumptions.
- Evaluate Critically:** Consider how feedback aligns with your goals and constraints. If you disagree, articulate your reasoning respectfully.
- Act Promptly:** Implement agreed-upon changes quickly. Supervisors lose confidence when students repeatedly ignore advice or delay action.
- Report Back:** In subsequent meetings, describe what you did with previous feedback, showing you value their input and close the feedback loop.

Class Activity and Part 2 Deliverable

Problem Framing Lab

This hands-on workshop walks you through systematic problem refinement using structured templates. You will apply CRISP-DM thinking and problem statement formulation to your chosen topic, receiving peer and instructor feedback that sharpens your focus before submitting your final problem statement.

Activity Steps (90 minutes)

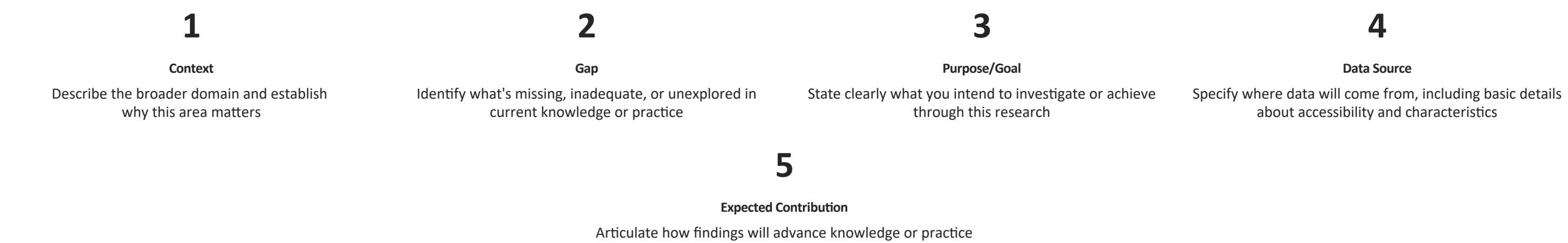


Part 2 Deliverable: Research Problem Statement

Due: End of the Week (submit via email)

Format: One-paragraph problem statement (150-250 words) plus brief supplementary information

Required Elements:



Assessment Criteria: Your problem statement will be evaluated on clarity (can readers understand exactly what you're studying?), feasibility (is this achievable in available time with accessible data?), significance (does it matter academically and practically?), and specificity (is it focused enough for deep investigation?).